ModelAudit — Summary (page 1)


```
{
  "dataset": "german",
  "model": "xgb",
  "scope": "research demonstration only; thresholds are local review heuristics",
  "evidence_sha256": "f227dc34e8a95f40d643b2cc33ad113d3485495bda17bb91e1f69a7da74b5792",
  "findings": [
    {
      "test_id": "CAL-ECE-01",
      "severity": "medium",
      "confirmation": "reproduced"
    },
    {
      "test_id": "STAB-CSI-01",
      "severity": "medium",
      "confirmation": "reproduced"
    },
    {
      "test_id": "FAIR-SEX-01",
      "severity": "medium",
      "confirmation": "not_reproduced"
    },
    {
      "test_id": "FAIR-AGE-01",
      "severity": "high",
      "confirmation": "reproduced"
    },
    {
      "test_id": "ROB-MONO-01",
      "severity": "high",
      "confirmation": "reproduced"
    }
  ]
}
```

ModelAudit — PERF-AUC-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "auc": 0.8032,
    "auc_ci95": [
      0.7542416490663165,
      0.8622310236060235
    ],
    "ks": 0.4780952380952381,
    "gini": 0.6064,
    "count": 250
  },
  "threshold": {
    "metric": "auc",
    "warn_below": 0.75,
    "fail_below": 0.7
  }
}
```

ModelAudit — CAL-ECE-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "brier": 0.15993010527814608,
    "ece": 0.04220368146896362,
    "calibration_in_the_large": -0.004029643535614025,
    "hosmer_lemeshow_stat": 4.943998601412295,
    "hosmer_lemeshow_df": 8,
    "hosmer_lemeshow_p": 0.7635445425409082,
    "reliability_bins": [
      {
        "bin": 0,
        "count": 25,
        "observed_rate": 0.0,
        "predicted_rate": 0.04980104923248291
      },
      {
        "bin": 1,
        "count": 25,
        "observed_rate": 0.08,
        "predicted_rate": 0.08432767868041992
      },
      {
        "bin": 2,
        "count": 25,
        "observed_rate": 0.04,
        "predicted_rate": 0.1335565662384033
      },
      {
        "bin": 3,
        "count": 25,
        "observed_rate": 0.2,
        "predicted_rate": 0.18302595138549804
      },
      {
        "bin": 4,
        "count": 25,
        "observed_rate": 0.2,
        "predicted_rate": 0.23446649551391602
      },
      {
        "bin": 5,
        "count": 25,
        "observed_rate": 0.32,
        "predicted_rate": 0.2953898811340332
      },
      {
        "bin": 6,
        "count": 25,
        "observed_rate": 0.44,
        "predicted_rate": 0.38573345184326174
      },
      {
        "bin": 7,
        "count": 25,
        "observed_rate": 0.44,
```

ModelAudit — CAL-ECE-01 (page 2)


```
        "predicted_rate": 0.45941967010498047
      },
      {
        "bin": 8,
        "count": 25,
        "observed_rate": 0.52,
        "predicted_rate": 0.5495951080322266
      },
      {
        "bin": 9,
        "count": 25,
        "observed_rate": 0.76,
        "predicted_rate": 0.66498046875
      }
    ]
  },
  "threshold": {
    "metric": "ece",
    "warn_above": 0.03,
    "fail_above": 0.05
  }
}
```

ModelAudit — STAB-PSI-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "psi": 0.08555588532453444,
    "baseline_test_psi": 0.04394032324453639,
    "scenario": "amount +25% and age +5"
  },
  "threshold": {
    "metric": "psi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

ModelAudit — STAB-CSI-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "max_csi": 0.12116270428342113,
    "feature": "A05",
    "baseline_test_csi": 0.05001226836672981
  },
  "threshold": {
    "metric": "max_csi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

ModelAudit — FAIR-SEX-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "threshold": 0.25,
    "groups": {
      "female": {
        "approval_rate": 0.42168674698795183,
        "good_outcome_tpr": 0.5660377358490566
      },
      "male_or_other": {
        "approval_rate": 0.5029940119760479,
        "good_outcome_tpr": 0.6311475409836066
      }
    },
    "demographic_parity_difference": 0.08130726498809604,
    "equalized_odds_difference": 0.06510980513455,
    "approval_rate_ratio": 0.8383534136546186,
    "overall_approval_rate": 0.476,
    "approved_observed_default_rate": 0.10084033613445378
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

ModelAudit — FAIR-AGE-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "threshold": 0.25,
    "groups": {
      "35plus": {
        "approval_rate": 0.5567010309278351,
        "good_outcome_tpr": 0.6842105263157895
      },
      "under35": {
        "approval_rate": 0.42483660130718953,
        "good_outcome_tpr": 0.5555555555555556
      }
    },
    "demographic_parity_difference": 0.13186442962064554,
    "equalized_odds_difference": 0.1286549707602339,
    "approval_rate_ratio": 0.7631324134592108,
    "overall_approval_rate": 0.476,
    "approved_observed_default_rate": 0.10084033613445378
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

ModelAudit — ROB-MONO-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "test_feature": "A02 loan duration months (stress proxy)",
    "violations": 5,
    "sample_rows": 250,
    "max_absolute_change": 0.23249408602714539,
    "missing_predictions_finite": true
  },
  "threshold": {
    "metric": "violations",
    "warn_above": 0,
    "fail_above": 1
  }
}
```

ModelAudit — DQ-CAT-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "rows": 250,
    "null_cells": 0,
    "duplicate_feature_rows": 0,
    "invalid_categories": 0,
    "label_out_of_range": 0,
    "id_and_target_excluded": true
  },
  "threshold": {
    "metric": "invalid_categories",
    "warn_above": 0,
    "fail_above": 100
  }
}
```

ModelAudit — FAIR-TRADEOFF (page 1)


```
{
  "status": "information",
  "values": {
    "thresholds": [
      {
        "threshold": 0.15,
        "groups": {
          "female": {
            "approval_rate": 0.21686746987951808,
            "good_outcome_tpr": 0.32075471698113206
          },
          "male_or_other": {
            "approval_rate": 0.32335329341317365,
            "good_outcome_tpr": 0.4262295081967213
          }
        },
        "demographic_parity_difference": 0.10648582353365557,
        "equalized_odds_difference": 0.10547479121558923,
        "approval_rate_ratio": 0.6706827309236948,
        "overall_approval_rate": 0.288,
        "approved_observed_default_rate": 0.041666666666666664
      },
      {
        "threshold": 0.2,
        "groups": {
          "female": {
            "approval_rate": 0.3373493975903614,
            "good_outcome_tpr": 0.4716981132075472
          },
          "male_or_other": {
            "approval_rate": 0.40718562874251496,
            "good_outcome_tpr": 0.5163934426229508
          }
        },
        "demographic_parity_difference": 0.06983623115215354,
        "equalized_odds_difference": 0.04469532941540366,
        "approval_rate_ratio": 0.8284904323175053,
        "overall_approval_rate": 0.384,
        "approved_observed_default_rate": 0.08333333333333333
      },
      {
        "threshold": 0.25,
        "groups": {
          "female": {
            "approval_rate": 0.42168674698795183,
            "good_outcome_tpr": 0.5660377358490566
          },
          "male_or_other": {
            "approval_rate": 0.5029940119760479,
            "good_outcome_tpr": 0.6311475409836066
          }
        },
        "demographic_parity_difference": 0.08130726498809604,
        "equalized_odds_difference": 0.06510980513455,
        "approval_rate_ratio": 0.8383534136546186,
        "overall_approval_rate": 0.476,
```

ModelAudit — FAIR-TRADEOFF (page 2)


```
        "approved_observed_default_rate": 0.10084033613445378
      },
      {
        "threshold": 0.3,
        "groups": {
          "female": {
            "approval_rate": 0.5421686746987951,
            "good_outcome_tpr": 0.7169811320754716
          },
          "male_or_other": {
            "approval_rate": 0.5688622754491018,
            "good_outcome_tpr": 0.6885245901639344
          }
        },
        "demographic_parity_difference": 0.02669360075030669,
        "equalized_odds_difference": 0.028456541911537236,
        "approval_rate_ratio": 0.9530754597336714,
        "overall_approval_rate": 0.56,
        "approved_observed_default_rate": 0.12857142857142856
      },
      {
        "threshold": 0.35,
        "groups": {
          "female": {
            "approval_rate": 0.5783132530120482,
            "good_outcome_tpr": 0.7547169811320755
          },
          "male_or_other": {
            "approval_rate": 0.6287425149700598,
            "good_outcome_tpr": 0.7459016393442623
          }
        },
        "demographic_parity_difference": 0.05042926195801167,
        "equalized_odds_difference": 0.04444444444444445,
        "approval_rate_ratio": 0.9197934595524957,
        "overall_approval_rate": 0.612,
        "approved_observed_default_rate": 0.1437908496732026
      }
    ]
  },
  "threshold": null
}
```

ModelAudit — EXPL-SHAP-01 (page 1)


```
{
  "status": "information",
  "values": {
    "dataset": "german",
    "rows_explained": 100,
    "xgb_shap_global_top10": [
      {
        "feature": "cat__A01_A14",
        "mean_abs_shap": 0.4713290333747864
      },
      {
        "feature": "num__A02",
        "mean_abs_shap": 0.235748752951622
      },
      {
        "feature": "cat__A03_A34",
        "mean_abs_shap": 0.20800858736038208
      },
      {
        "feature": "num__A05",
        "mean_abs_shap": 0.19495922327041626
      },
      {
        "feature": "cat__A01_A11",
        "mean_abs_shap": 0.18921680748462677
      },
      {
        "feature": "num__A08",
        "mean_abs_shap": 0.1332424283027649
      },
      {
        "feature": "num__A13",
        "mean_abs_shap": 0.1268044263124466
      },
      {
        "feature": "cat__A04_A40",
        "mean_abs_shap": 0.11415299773216248
      },
      {
        "feature": "cat__A07_A74",
        "mean_abs_shap": 0.1027626022696495
      },
      {
        "feature": "cat__A04_A41",
        "mean_abs_shap": 0.10129211097955704
      }
    ],
    "xgb_first_case_adverse_top3": [
      {
        "feature": "cat__A01_A14",
        "shap_log_odds": 0.31417495012283325
      },
      {
        "feature": "num__A05",
        "shap_log_odds": 0.15592725574970245
      },
```

ModelAudit — EXPL-SHAP-01 (page 2)


```
      {
        "feature": "num__A08",
        "shap_log_odds": 0.11538189649581909
      }
    ],
    "scorecard_first_case_reason_codes": [
      "A01",
      "A08",
      "A06"
    ],
    "scorecard_top3_overlap_under_1pct_amount_change": 1.0,
    "warning": "SHAP is model explanation in log-odds space, not a causal reason or lawful adverse-action
  notice."
  },
  "threshold": null
}
```